

Reform AI — Style Implementation Recommendations Document

Based on: AI Visualization Tool Smoke Test and Final Style System

Prepared for: Product, Engineering, and Design Teams

Objective: Define the optimal implementation strategy, prioritization, and ordering of design styles to ensure spatial consistency, product reliability, and demo readiness.

1. Guiding Principle for Style Implementation

The implementation of styles within the AI Visualization Tool must be governed by the following core principles:

1.1 Spatial Preservation (Critical Requirement)

All generated outputs must strictly preserve the spatial characteristics of the base image, including:

- Camera angle
- Room geometry and proportions
- Structural elements (walls, windows, layout)
- Background continuity

This is the most critical requirement, as multiple failure cases identified in the smoke test were directly related to violations of spatial integrity.

1.2 Controlled Transformation (Not Scene Regeneration)

The system must perform controlled transformations rather than full scene regeneration. Styles should enhance the existing scene without:

- Overwriting environmental context
 - Reconstructing geometry
 - Replacing key structural elements
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1.3 Visual Coherence Over Stylistic Extremes

Consistency and realism must be prioritized over aggressive stylistic expression, particularly in production and demo environments.

2. Cross-Referenced Style Classification

Based on the intersection of smoke test performance and the finalized style system, styles are categorized into three tiers:

2.1 Tier 1 — Core Production / Demo-Safe Styles

These styles demonstrate high reliability, strong spatial preservation, and consistent visual outputs, making them highly relevant for production:

- Industrial
- Japanese*
- Minimalist
- Japandi
- Contemporary
- Biophilic
- Modern
- Rustic

**Recommendation Note: Given potential obsolescence of the "Japanese" style for implementation in Colombia, it is recommended to exclude it from the final implementation list, keeping only "Japandi".*

Characteristics:

- Preserve geometry and camera angle
- Maintain window and background integrity
- Produce stable, coherent outputs across all test phases

These styles should form the foundation of the production system.

2.2 Tier 2 — Controlled Use / Secondary Styles

These styles are usable under controlled conditions but may introduce minor inconsistencies:

- Vintage
- Industrial
- Coastal
- Bohemian
- Neoclassic

Observed Risks:

- Minor structural or furniture variations
- Increased stylistic interpretation
- Slight deviations in object geometry

These styles should be implemented with constraints and validation layers.

2.3 Tier 3 — Not Production Ready

These styles introduce significant inconsistencies and should not be included in production environments:

- Farmhouse
- Mid-Century Modern
- French Country
- Ski Chalet
- Cyberpunk
- Medieval

Common Failure Patterns:

- Window obstruction or modification
- Background replacement
- Camera angle shifts
- Loss of spatial coherence

These styles require substantial refinement before being considered for production use.

3. Recommended Implementation Order

A phased rollout strategy is recommended to ensure stability and controlled expansion:

Phase 1 — Core Stable Styles (Go-to-Market)

1. Minimalist
2. Industrial
3. Japanese*
4. Japandi
5. Contemporary
6. Modern

Objective:

Establish a highly reliable baseline with predictable and demo-ready outputs.

Phase 2 — Controlled Expansion

6. Biophilic
7. Rustic
8. Vintage

Objective:

Expand stylistic diversity while maintaining system stability.

Phase 3 — Experimental / Beta

10. Coastal
11. Bohemian
12. Neoclassic

Objective:

Test broader stylistic flexibility under controlled conditions with validation mechanisms.

Phase 4 — Deferred (Do Not Enable Yet)

- Cyberpunk
- Ski Chalet
- Medieval
- French Country
- Farmhouse

Reason:

These styles currently fail to meet minimum requirements for spatial consistency and reliability.

4. Technical Implementation Guidelines

4.1 Constraint Layer (Mandatory)

A constraint system must be implemented across all styles to enforce:

- Camera angle locking
- Geometry preservation
- Window and structural integrity
- Background consistency

This layer is essential to prevent the most critical failure modes identified in testing.

4.2 Style Intensity Control

Each style should support adjustable transformation levels:

- Low (default for production)
- Medium
- High

Production environments should default to **low intensity** to ensure stability.

4.3 Window Handling Rules (Critical)

The system must enforce strict rules regarding windows:

- Do not replace exterior backgrounds
- Do not obstruct or modify windows
- Do not alter window size or position

Window handling is the most consistent failure point across all test phases and must be addressed systemically.

4.4 Furniture Preservation Layer

When furniture is present, the system must:

- Preserve object structure and identity
 - Avoid deformation, particularly in key elements such as headboards
 - Maintain accurate placement and scale
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5. Alignment with the Style System

The finalized style system defines a standardized taxonomy of styles with consistent naming conventions (English and Spanish).

Recommendation:

- Use this system as the single source of truth
- Map each style to:
 - Tier classification
 - Risk level
 - Constraint profile

This ensures consistency across product, engineering, and design.

6. Product and UX Recommendations

6.1 Demo Mode (Recommended)

A curated “Demo Mode” should be implemented with:

- Only Tier 1 styles
- Selected Tier 2 styles (validated)

This ensures predictable, high-quality outputs in all demo scenarios.

6.2 Advanced Mode

An advanced mode may expose additional styles with clear labeling:

- “Experimental”
- “May alter layout”

This allows flexibility without compromising user expectations.

7. Spatial Coherence Requirement

A strict system requirement must be defined as follows:

All style outputs must achieve 100 percent spatial coherence with the base image.

This includes:

- Identical perspective
- Consistent geometry
- Preservation of structural elements
- Environmental continuity

This requirement is currently not met across several styles and must be enforced through system constraints.

8. Strategic Conclusion

Not all styles should be treated equally within the system.

The implementation must be:

- Curated
- Tiered
- Constraint-driven

Key Insight:

Product success depends more on spatial consistency and reliability than on stylistic diversity.

9. Immediate Recommended Actions

1. Deploy Tier 1 styles as default production options
2. Implement a geometry and constraint enforcement system

3. Disable Tier 3 styles in production environments
4. Introduce automated spatial coherence validation
5. Define and deploy a controlled Demo Mode

10. Disclaimer Recommendation

We recommend adding the following disclaimer after every image/visualization generation: "AI images may generate errors and hallucinations. Before starting a project or purchasing any product, we recommend verifying with a professional in our network and/or checking the corresponding measurements and legal requirements for any type of renovation."

Final Assessment:

The AI Visualization Tool is conditionally demo-ready and has strong potential as a core product differentiator. Achieving full production reliability will depend on enforcing spatial constraints, improving consistency across styles, and implementing a structured, tiered style system.
